

Database Assignment 1

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Note : Use Emp, dept and salgrade table

1. To list all records with sal > 2000 and comm>200

```
mysql> select *from emp;
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME | JOB      | MGR | HIREDATE | SAL      | COMM      | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7369 | SMITH | CLERK    | 7902 | 1980-12-17 | 800.00 | NULL      | 20      |
| 7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00    | 30      |
| 7521 | WARD  | SALESMAN | 7698 | 1981-02-22 | 1250.00 | 500.00    | 30      |
| 7566 | JONES | MANAGER  | 7839 | 1981-04-02 | 2975.00 | NULL      | 20      |
| 7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00 | 1400.00   | 30      |
| 7698 | BLAKE | MANAGER  | 7839 | 1981-05-01 | 2850.00 | NULL      | 30      |
| 7782 | CLARK | MANAGER  | 7839 | 1981-06-09 | 2450.00 | NULL      | 10      |
| 7788 | SCOTT | ANALYST  | 7566 | 1982-12-09 | 3000.00 | NULL      | 20      |
| 7839 | KING  | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL      | 10      |
| 7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 | 0.00      | 30      |
| 7876 | ADAMS | CLERK    | 7788 | 1983-01-12 | 1100.00 | NULL      | 20      |
| 7900 | JAMES | CLERK    | 7698 | 1981-12-03 | 950.00  | NULL      | 30      |
| 7902 | FORD  | ANALYST  | 7566 | 1981-12-03 | 3000.00 | NULL      | 20      |
| 7934 | MILLER | CLERK    | 7782 | 1982-01-23 | 1300.00 | NULL      | 10      |
+-----+-----+-----+-----+-----+-----+-----+-----+
14 rows in set (0.00 sec)

mysql> select * from emp where sal>2000 and comm>2001
-> ;
Empty set (0.01 sec)

mysql> select * from emp where sal>2000 and comm>200;
Empty set (0.00 sec)
```

2. To list all record with job='Clerk' or sal>2000

```
mysql> select * from emp where job='clerk' or sal>2000;
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME | JOB      | MGR | HIREDATE | SAL      | COMM      | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7369 | SMITH | CLERK    | 7902 | 1980-12-17 | 800.00 | NULL      | 20      |
| 7566 | JONES | MANAGER  | 7839 | 1981-04-02 | 2975.00 | NULL      | 20      |
| 7698 | BLAKE | MANAGER  | 7839 | 1981-05-01 | 2850.00 | NULL      | 30      |
| 7782 | CLARK | MANAGER  | 7839 | 1981-06-09 | 2450.00 | NULL      | 10      |
| 7788 | SCOTT | ANALYST  | 7566 | 1982-12-09 | 3000.00 | NULL      | 20      |
| 7839 | KING  | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL      | 10      |
| 7876 | ADAMS | CLERK    | 7788 | 1983-01-12 | 1100.00 | NULL      | 20      |
| 7900 | JAMES | CLERK    | 7698 | 1981-12-03 | 950.00  | NULL      | 30      |
| 7902 | FORD  | ANALYST  | 7566 | 1981-12-03 | 3000.00 | NULL      | 20      |
| 7934 | MILLER | CLERK    | 7782 | 1982-01-23 | 1300.00 | NULL      | 10      |
+-----+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.01 sec)
```

3. To list all the record with sal=1250 or 1100 or 2850

```
mysql> select * from emp where sal in (1250,1100,2850);
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20

```
4 rows in set (0.00 sec)
```

4. To list all employees with sal>1250 and <2850

```
mysql> select * from emp where sal between '1250' and '2850';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

```
7 rows in set (0.00 sec)
```

5. To list all employees with name ends with AS

```
mysql> select * from emp where ename like '%AS';
```

Empty set (0.03 sec)

```
mysql> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

```
mysql> select * from emp where ename REGEXP 'AS$';
Empty set (0.09 sec)
```

6. To list all employees with job starts with C and ends with K

```
mysql> select * from emp where job like 'C%K';
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME  | JOB   | MGR  | HIREDATE   | SAL      | COMM  | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7369  | SMITH  | CLERK | 7902 | 1980-12-17 | 800.00   | NULL  | 20     |
| 7876  | ADAMS  | CLERK | 7788 | 1983-01-12 | 1100.00  | NULL  | 20     |
| 7900  | JAMES  | CLERK | 7698 | 1981-12-03 | 950.00   | NULL  | 30     |
| 7934  | MILLER | CLERK | 7782 | 1982-01-23 | 1300.00  | NULL  | 10     |
+-----+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> select * from emp where job REGEXP '^C.*K$';
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME  | JOB   | MGR  | HIREDATE   | SAL      | COMM  | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7369  | SMITH  | CLERK | 7902 | 1980-12-17 | 800.00   | NULL  | 20     |
| 7876  | ADAMS  | CLERK | 7788 | 1983-01-12 | 1100.00  | NULL  | 20     |
| 7900  | JAMES  | CLERK | 7698 | 1981-12-03 | 950.00   | NULL  | 30     |
| 7934  | MILLER | CLERK | 7782 | 1982-01-23 | 1300.00  | NULL  | 10     |
+-----+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

7. To list all employees with job contains L at third position and M at third last position

```
mysql> SELECT * FROM EMP WHERE JOB LIKE '__L%M__';
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL      | COMM  | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7499  | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00  | 300.00 | 30     |
| 7521  | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250.00  | 500.00 | 30     |
| 7654  | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00  | 1400.00 | 30     |
| 7844  | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00  | 0.00   | 30     |
+-----+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM EMP WHERE JOB REGEXP '^..L.*M..';
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL      | COMM  | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7499  | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00  | 300.00 | 30     |
| 7521  | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250.00  | 500.00 | 30     |
| 7654  | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00  | 1400.00 | 30     |
| 7844  | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00  | 0.00   | 30     |
+-----+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

8. To list all the record with sal not equal to 1250 or 1100 or 2850

```
mysql> SELECT * FROM EMP WHERE SAL NOT IN (1250,1100,2850);
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

10 rows in set (0.00 sec)

9. To list all employees with salnot >1250 and <2850

```
mysql> SELECT * FROM EMP WHERE SAL NOT BETWEEN '1250' AND '2850';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20

7 rows in set (0.00 sec)

10. To list all employees with job starts with C , E at 3rd position and ends with K

```
mysql> SELECT * FROM EMP WHERE JOB LIKE 'C_E%K';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

4 rows in set (0.00 sec)

```
mysql> SELECT * FROM EMP WHERE JOB REGEXP '^C.E.*K$'
-> ;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

4 rows in set (0.00 sec)

11. To list all rows with comm is null

```
mysql> SELECT * FROM EMP WHERE COMM IS NULL;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

10 rows in set (0.00 sec)

12. To list all employees with sal is null and name starts with 'S'

```
mysql> SELECT * FROM EMP WHERE SAL IS NULL AND ENAME LIKE 'S%';
Empty set (0.00 sec)

mysql> SELECT * FROM EMP WHERE SAL IS NULL AND ENAME REGEXP '^S';
Empty set (0.00 sec)
```

13. To list all employees with job contains 5 characters

```
mysql> SELECT * FROM EMP WHERE JOB REGEXP '^.{5}$';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM EMP WHERE JOB REGEXP '^.....$';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM EMP WHERE JOB LIKE '_____';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

```
4 rows in set (0.00 sec)
```

14. To list all employees with name contain 'A' at 1 position and job Contains 5 characters

```
mysql> SELECT * FROM EMP WHERE ENAME LIKE 'A%' AND JOB LIKE '_____';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM EMP WHERE ENAME REGEXP '^A' AND JOB REGEXP '^.{5}$';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20

```
1 row in set (0.00 sec)
```

Q2. Solve the following

1. Retrieve the details (Name, Salary and dept no) of the emp who are working in department code 20, 30 and 40.

```
mysql>
mysql> SELECT ENAME,SAL,DEPTNO FROM EMP WHERE DEPTNO IN (20,30,40);
```

ENAME	SAL	DEPTNO
SMITH	800.00	20
ALLEN	1600.00	30
WARD	1250.00	30
JONES	2975.00	20
MARTIN	1250.00	30
BLAKE	2850.00	30
SCOTT	3000.00	20
TURNER	1500.00	30
ADAMS	1100.00	20
JAMES	950.00	30
FORD	3000.00	20

```
11 rows in set (0.00 sec)
```

2. Display the total salary of all employees . Total salary will be calculated as $sal + comm + sal * 0.10$

```
mysql> SELECT SUM(SAL+COMM+SAL*0.10) AS TOTAL_SAL FROM EMP;
```

TOTAL_SAL
8360.0000

```
1 row in set (0.00 sec)
```

2. List the Name and job of the emp who have joined before 1 jan 1986 and whose salary range is between 1200 and 2500. Display the columns with user defined Column headers.

```
mysql> select ename 'Emp_name',job,sal 'Emp_sal' from emp where hiredate<'1986-01-01'and sal between 1200 and 2500;
```

Emp_name	job	Emp_sal
ALLEN	SALESMAN	1600.00
WARD	SALESMAN	1250.00
MARTIN	SALESMAN	1250.00
CLARK	MANAGER	2450.00
TURNER	SALESMAN	1500.00
MILLER	CLERK	1300.00

```
6 rows in set (0.00 sec)
```

```
mysql> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

- List the empno, name, and department number of the emp works under manager with id 7698

```
mysql> select empno,name,deptno from emp where mgr=7698;
ERROR 1054 (42S22): Unknown column 'name' in 'field list'
mysql> select empno,ename,deptno from emp where mgr=7698;
```

empno	ename	deptno
7499	ALLEN	30
7521	WARD	30
7654	MARTIN	30
7844	TURNER	30
7900	JAMES	30

```
5 rows in set (0.00 sec)
```

- List the name, job, and salary of the emp who are working in departments 10 and 30.

```
mysql> select ename,job,deptno,sal from emp where deptno in (10,30);
```

ename	job	deptno	sal
ALLEN	SALESMAN	30	1600.00
WARD	SALESMAN	30	1250.00
MARTIN	SALESMAN	30	1250.00
BLAKE	MANAGER	30	2850.00
CLARK	MANAGER	10	2450.00
KING	PRESIDENT	10	5000.00
TURNER	SALESMAN	30	1500.00
JAMES	CLERK	30	950.00
MILLER	CLERK	10	1300.00

```
9 rows in set (0.00 sec)
```

-

5. Display name concatenated with dept code separated by comma and space. Name the column as 'Emp info'.

```
mysql> select concat(ename,' , ',deptno) 'Emp_info' from emp;
```

Emp_info
SMITH , 20
ALLEN , 30
WARD , 30
JONES , 20
MARTIN , 30
BLAKE , 30
CLARK , 10
SCOTT , 20
KING , 10
TURNER , 30
ADAMS , 20
JAMES , 30
FORD , 20
MILLER , 10

14 rows in set (0.00 sec)

5. Display the emp details who do not have manager.

```
mysql> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

14 rows in set (0.00 sec)

```
mysql> select * from emp where mgr is null;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10

1 row in set (0.00 sec)

6.

7. Write a query which will display name, department no and date of joining of all employee who were joined January 1, 1981 and March 31, 1983. Sort it based on date of joining (ascending).

```
mysql> select ename,deptno,hiredate from emp where hiredate between '1981-01-01' and '1983-03-31' order by hiredate;
```

ename	deptno	hiredate
ALLEN	30	1981-02-20
WARD	30	1981-02-22
JONES	20	1981-04-02
BLAKE	30	1981-05-01
CLARK	10	1981-06-09
TURNER	30	1981-09-08
MARTIN	30	1981-09-28
KING	10	1981-11-17
JAMES	30	1981-12-03
FORD	20	1981-12-03
MILLER	10	1982-01-23
SCOTT	20	1982-12-09
ADAMS	20	1983-01-12

13 rows in set (0.00 sec)

8. Display the employee details where the job contains word 'AGE' anywhere in the Job

```
mysql> select * from emp where job like '%AGE%';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10

3 rows in set (0.00 sec)

```
mysql> select * from emp where job regexp '^. *age.*$';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10

2 rows in set (0.06 sec)

11. List the details of the employee , whose names start with 'A' and end with 'S' or
whose names contains N as the second or third character, and ending with either 'N' or 'S'.

```
mysql> select * from emp where ename regexp '^a.*s$|^..n.*[ns]$|^..n.*[ns]$';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20

2 rows in set (0.00 sec)

12. List the names of the emp having '_' character in their name.

```
mysql> insert into emp values(7781,'tanu_dbda','analyst',7902,'2024-10-08',3000,300,20);
Query OK, 1 row affected (0.04 sec)

mysql> select *from emp;
+-----+-----+-----+-----+-----+-----+-----+-----+
| EMPNO | ENAME   | JOB       | MGR  | HIREDATE | SAL      | COMM   | DEPTNO |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7369 | SMITH   | CLERK     | 7902 | 1980-12-17 | 800.00   | NULL   | 20     |
| 7499 | ALLEN   | SALESMAN  | 7698 | 1981-02-20 | 1600.00  | 300.00 | 30     |
| 7521 | WARD    | SALESMAN  | 7698 | 1981-02-22 | 1250.00  | 500.00 | 30     |
| 7566 | JONES   | MANAGER   | 7839 | 1981-04-02 | 2975.00  | NULL   | 20     |
| 7654 | MARTIN  | SALESMAN  | 7698 | 1981-09-28 | 1250.00  | 1400.00 | 30     |
| 7698 | BLAKE   | MANAGER   | 7839 | 1981-05-01 | 2850.00  | NULL   | 30     |
| 7780 | jay_dbda | analyst   | 7902 | 2024-10-08 | 3000.00  | 300.00 | 20     |
| 7781 | tanu_dbda | analyst   | 7902 | 2024-10-08 | 3000.00  | 300.00 | 20     |
| 7782 | CLARK   | MANAGER   | 7839 | 1981-06-09 | 2450.00  | NULL   | 10     |
| 7788 | SCOTT   | ANALYST   | 7566 | 1982-12-09 | 3000.00  | NULL   | 20     |
| 7839 | KING    | PRESIDENT | NULL | 1981-11-17 | 5000.00  | NULL   | 10     |
| 7844 | TURNER  | SALESMAN  | 7698 | 1981-09-08 | 1500.00  | 0.00   | 30     |
| 7876 | ADAMS   | CLERK     | 7788 | 1983-01-12 | 1100.00  | NULL   | 20     |
| 7900 | JAMES   | CLERK     | 7698 | 1981-12-03 | 950.00   | NULL   | 30     |
| 7902 | FORD    | ANALYST   | 7566 | 1981-12-03 | 3000.00  | NULL   | 20     |
| 7934 | MILLER  | CLERK     | 7782 | 1982-01-23 | 1300.00  | NULL   | 10     |
+-----+-----+-----+-----+-----+-----+-----+-----+
16 rows in set (0.00 sec)

mysql> select ename from emp where ename like '%\_%';
+-----+
| ename   |
+-----+
| jay_dbda |
| tanu_dbda |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> select ename from emp where ename regexp '^.*_.*$';
+-----+
| ename   |
+-----+
| jay_dbda |
| tanu_dbda |
+-----+
2 rows in set (0.00 sec)
```

Single Row functions

1. To list all employees and their email, to generate email use 2 to 5 characters from ename
Concat it with 2 to 4 characters in job and then concat it with '@mycompany.com'

```
mysql> select ename,concat(substring(ename,1,5),substring(job,1,4),'@mycompany.com') as email from emp;
```

ename	email
SMITH	SMITHCLER@mycompany.com
ALLEN	ALLENSALE@mycompany.com
WARD	WARDSALE@mycompany.com
JONES	JONESMANA@mycompany.com
MARTIN	MARTISALE@mycompany.com
BLAKE	BLAKEMANA@mycompany.com
jay_dbda	jay_danal@mycompany.com
tanu_dbda	tanu_anal@mycompany.com
CLARK	CLARKMANA@mycompany.com
SCOTT	SCOTTANAL@mycompany.com
KING	KINGPRES@mycompany.com
TURNER	TURNESALE@mycompany.com
ADAMS	ADAMSCLER@mycompany.com
JAMES	JAMESCLER@mycompany.com
FORD	FORDANAL@mycompany.com
MILLER	MILLECLER@mycompany.com

```
16 rows in set (0.01 sec)
```

2. List all employees who joined in September.

```
mysql> select ename from emp where hiredate=9;
Empty set, 1 warning (0.00 sec)

mysql> select ename from emp where month(hiredate)=9;
```

ename
MARTIN
TURNER

```
2 rows in set (0.01 sec)
```

3. List the empno, name, and department number of the emp who have experience of 18 or more years and sort them based on their experience.

```
mysql> select empno,ename,deptno,floor(datediff(curdate(),hiredate)/365)
-> expirance from emp
-> where floor(datediff(curdate(),hiredate)/365>=18)
-> order by expirance;
```

empno	ename	deptno	expirance
7788	SCOTT	20	41
7876	ADAMS	20	41
7654	MARTIN	30	42
7698	BLAKE	30	42
7782	CLARK	10	42
7839	KING	10	42
7844	TURNER	30	42
7900	JAMES	30	42
7902	FORD	20	42
7934	MILLER	10	42
7369	SMITH	20	43
7499	ALLEN	30	43
7521	WARD	30	43
7566	JONES	20	43

14 rows in set (0.00 sec)

4. Display the employee details who joined on 3rd of any month or any year

```
mysql> select * from emp where day(hiredate)=3;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20

2 rows in set (0.00 sec)

5. display all employees who joined between years 1981 to 1983.

```
mysql> select * from emp where hiredate between '1981-01-01' and '1983-12-1';
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1982-12-09	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1983-01-12	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

13 rows in set (0.00 sec)

Group functions

6. Display the Highest, Lowest, Total & Average salary of all employee. Label the columns Maximum, Minimum, Total and Average respectively for each Department. Also round the result to the nearest whole number.

```
mysql> select sum(sal),round(sal),avg(sal),min(sal),max(sal) from emp;
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #2 of SELECT list contains
nonaggregated column 'iacsd0324.emp.SAL'; this is incompatible with sql_mode=only_full_group_b
y
mysql> select sum(sal),round(sal),avg(sal),min(sal),max(sal) from emp group by sal;
+-----+-----+-----+-----+-----+
| sum(sal) | round(sal) | avg(sal) | min(sal) | max(sal) |
+-----+-----+-----+-----+-----+
| 800.00 | 800 | 800.000000 | 800.00 | 800.00 |
| 1600.00 | 1600 | 1600.000000 | 1600.00 | 1600.00 |
| 2500.00 | 1250 | 1250.000000 | 1250.00 | 1250.00 |
| 2975.00 | 2975 | 2975.000000 | 2975.00 | 2975.00 |
| 2850.00 | 2850 | 2850.000000 | 2850.00 | 2850.00 |
| 12000.00 | 3000 | 3000.000000 | 3000.00 | 3000.00 |
| 2450.00 | 2450 | 2450.000000 | 2450.00 | 2450.00 |
| 5000.00 | 5000 | 5000.000000 | 5000.00 | 5000.00 |
| 1500.00 | 1500 | 1500.000000 | 1500.00 | 1500.00 |
| 1100.00 | 1100 | 1100.000000 | 1100.00 | 1100.00 |
| 950.00 | 950 | 950.000000 | 950.00 | 950.00 |
| 1300.00 | 1300 | 1300.000000 | 1300.00 | 1300.00 |
+-----+-----+-----+-----+-----+
```

```
mysql> select sum(sal) 'SAL_EMP',round(sal),avg(sal) 'AVERAGE_SAL',min(sal) 'MINIMUM',max(sal)
'MAXIMUM' from emp group by sal;
+-----+-----+-----+-----+-----+
| SAL_EMP | round(sal) | AVERAGE_SAL | MINIMUM | MAXIMUM |
+-----+-----+-----+-----+-----+
| 800.00 | 800 | 800.000000 | 800.00 | 800.00 |
| 1600.00 | 1600 | 1600.000000 | 1600.00 | 1600.00 |
| 2500.00 | 1250 | 1250.000000 | 1250.00 | 1250.00 |
| 2975.00 | 2975 | 2975.000000 | 2975.00 | 2975.00 |
| 2850.00 | 2850 | 2850.000000 | 2850.00 | 2850.00 |
| 12000.00 | 3000 | 3000.000000 | 3000.00 | 3000.00 |
| 2450.00 | 2450 | 2450.000000 | 2450.00 | 2450.00 |
| 5000.00 | 5000 | 5000.000000 | 5000.00 | 5000.00 |
| 1500.00 | 1500 | 1500.000000 | 1500.00 | 1500.00 |
| 1100.00 | 1100 | 1100.000000 | 1100.00 | 1100.00 |
| 950.00 | 950 | 950.000000 | 950.00 | 950.00 |
| 1300.00 | 1300 | 1300.000000 | 1300.00 | 1300.00 |
+-----+-----+-----+-----+-----+
12 rows in set (0.00 sec)
```

7. Display Department no and number of managers working in that department. Label the column as 'Total Number of Managers' for each department.

```
mysql> select deptno ,count(*) from emp where job='manager' group by deptno;
+-----+-----+
| deptno | count(*) |
+-----+-----+
| 20 | 1 |
| 30 | 1 |
| 10 | 1 |
+-----+-----+
3 rows in set (0.00 sec)
```

8. Get the Department number, and sum of Salary of all non managers where the sum is greater than 20000.

9.

```
mysql> select deptno ,sum(sal) from emp where job !='manager' group by deptno having sum(sal)>20000;  
Empty set (0.00 sec)
```